Storage Lab: Storage Queue Depth & Disk Utilization Metric Anomaly Detection

Duration : 30 Min.

Scenario

A web application is intermittently timing out. The server monitoring dashboard shows CPU at 25% and Memory at 40%, but disk response time is degraded. The engineer must use iostat -xz 1 10 to gather real-time disk statistics, parse %util (disk saturation), average queue size (avgqu-sz), and write await times, determine whether the storage device is experiencing I/O queuing, and generate a metrics summary.

Task Objectives

Perform the following storage latency and queue depth analysis tasks inside your `student_workspace` directory:

> **Pre-Staged Files in `student_workspace/`:** - **`iostat_capture.txt`**: Real-time disk I/O metrics captured via `iostat -xz`. - **`iostat_analysis.json`**: Starter diagnostic analysis JSON with `TODO_*` placeholders.

1. Inspect `iostat_capture.txt` to parse device utilization, average queue depth, and latency breakdowns.
2. Evaluate whether device utilization indicates storage controller saturation.
3. Differentiate between read and write latency components to isolate the primary I/O bottleneck.
4. Record your diagnostic metrics and saturation verdict in `iostat_analysis.json`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Command execution: iostat -xz correctly invoked with appropriate sampling interval and count	2 Marks
TC2	Utilization parsing: %util extracted accurately and compared against saturation baseline	2 Marks
TC3	Queue depth evaluation: avgqu-sz correctly interpreted indicating queued I/O requests	2 Marks
TC4	Latency breakdown: r_await and w_await correctly extracted to isolate write-heavy bottleneck	2 Marks
TC5	Summary report: iostat_analysis.json contains structured metrics and clear bottleneck conclusion	2 Marks

Total Score: 10 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.